

Logical Bilateralism - Proofs, Models and Applications

Session 5 – Recent developments and further questions to ask

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Today's session

Fun with bilateralism!

- Bilateralism as a road to contra-classical logics
- Do assertion/proof and denial/refutation need to be (in)dependent?
- Nice results to obtain: Uniqueness in bilateral frameworks

Bilateralism as a road to contra-classical logics

Background: Connexive Logics

Connexive Logics

- Belong to *contra-classical* logics (Humberstone 2000):
 - Contain theorems that are non-theorems of CL
 - Other examples: Abelian logics containing the axiom schema $((A \rightarrow B) \rightarrow B) \rightarrow A$, Arieli & Avron's (1996) bilattice logic
- Name implies that there must be some *connection* between antecedent and consequent of valid implications or, resp., between premises and conclusions of valid inferences
- Basic idea: no formula provably implies or is implied by its own negation

Background: Connexive Logics

Most common definition:

A logic is connexive iff the system validates

1.) Aristotle's Theses:

- AT: $\sim (\sim A \rightarrow A)$
- AT': $\sim (A \rightarrow \sim A)$

2.) Boethius' Theses:

- BT: $(A \rightarrow B) \rightarrow \sim (A \rightarrow \sim B)$
- BT': $(A \rightarrow \sim B) \rightarrow \sim (A \rightarrow B)$

3.) satisfies non-symmetry of implication, i.e., $(A \rightarrow B) \rightarrow (B \rightarrow A)$ is not a theorem

Two approaches to connexive systems

Approaches to Connexivity

1. Refining the **truth** conditions for implication
 - ⇒ Angell (1962)/McCall (1966) (material implication + tweak)
 - ⇒ Routley (1978) (relevant implication + star negation)
2. Refining the **falsity** conditions for implication
 - ⇒ Wansing's (2005) system C: connexive variant of Nelson's constructive logic N4

Connexive implication - proof-theoretically understood

Natural deduction system for N4:

$$\frac{A}{\sim\sim A} \sim\sim I \quad \frac{\sim\sim A}{A} \sim\sim E$$

$$\frac{\sim A}{\sim (A \wedge B)} \sim\wedge I_1 \quad \frac{\sim B}{\sim (A \wedge B)} \sim\wedge I_2 \quad \frac{\sim (A \wedge B) \quad \begin{array}{c} [\sim A] \\ \vdots \\ \overline{C} \end{array} \quad \begin{array}{c} [\sim B] \\ \vdots \\ \overline{C} \end{array}}{C} \sim\wedge E$$

$$\frac{\begin{array}{c} [A] \\ \vdots \\ \overline{B} \end{array}}{A \rightarrow B} \rightarrow I \quad \frac{A \quad A \rightarrow B}{B} \rightarrow E \quad \frac{A \quad \sim B}{\sim (A \rightarrow B)} \rightarrow E \quad \frac{\sim (A \rightarrow B)}{A} \sim\rightarrow I \quad \frac{\sim (A \rightarrow B)}{\sim B} \sim\rightarrow E_1 \quad \frac{\sim (A \rightarrow B)}{\sim B} \sim\rightarrow E_2$$

Connexive implication - proof-theoretically understood

Natural deduction system for N4:

$$\frac{A}{\sim\sim A} \sim\sim I \quad \frac{\sim\sim A}{A} \sim\sim E$$

$$\frac{\sim A}{\sim (A \wedge B)} \sim\wedge I_1 \quad \frac{\sim B}{\sim (A \wedge B)} \sim\wedge I_2 \quad \frac{\sim (A \wedge B) \quad \begin{array}{c} [\sim A] \\ \vdots \\ \overline{C} \end{array} \quad \begin{array}{c} [\sim B] \\ \vdots \\ \overline{C} \end{array}}{C} \sim\wedge E$$

$$\frac{\begin{array}{c} [A] \\ \vdots \\ \overline{B} \end{array}}{A \rightarrow B} \rightarrow I \quad \frac{A \quad A \rightarrow B}{B} \rightarrow E \quad \frac{A \quad \sim B}{\sim (A \rightarrow B)} \rightarrow E \quad \frac{\sim (A \rightarrow B) \quad \sim B}{A} \sim\rightarrow I \quad \frac{\sim (A \rightarrow B)}{\sim B} \sim\rightarrow E_1 \quad \frac{\sim (A \rightarrow B)}{\sim B} \sim\rightarrow E_2$$

Connexive implication - proof-theoretically understood

From N4...

$$\begin{array}{c}
 [A] \\
 \vdots \\
 \overline{B} \\
 \hline
 A \rightarrow B
 \end{array}
 \rightarrow_I
 \frac{A \quad A \rightarrow B}{B}
 \rightarrow_E
 \frac{A \quad \sim B}{\sim(A \rightarrow B)}
 \sim \rightarrow_I
 \frac{\sim(A \rightarrow B) \quad A}{\sim(A \rightarrow B)}
 \sim \rightarrow_{E_1}
 \frac{\sim(A \rightarrow B)}{\sim B}
 \sim \rightarrow_{E_2}$$

...to C:

$$\begin{array}{c}
 [A] \\
 \vdots \\
 \overline{B} \\
 \hline
 A \rightarrow B
 \end{array}
 \rightarrow_I
 \frac{A \quad A \rightarrow B}{B}
 \rightarrow_E
 \frac{[A] \quad \vdots \quad \overline{\sim B}}{\sim(A \rightarrow B)}
 \sim \rightarrow_I
 \frac{A \quad \sim(A \rightarrow B)}{\sim B}
 \sim \rightarrow_E$$

From bi-intuitionistic to bi-connexive logic: changing refutation conditions of implication

López-Escobar's dual version of BHK clauses: logical connectives are interpreted in terms of a primitive and constructive notion of *refutation*.

Dual/Bi-connexive logic:

- i)^d A refutation of $A \wedge B$ is a pair $\langle i, d \rangle$, where $i = 0$ and d is a refutation of A , or $i = 1$ and d is a refutation of B .
- ii)^d A refutation of $A \vee B$ is a pair $\langle d, e \rangle$, where d is a refutation of A and e is a refutation of B .
- iii)^d_c A refutation of $A \rightarrow B$ is a function that maps any proof of A to a refutation of B .
- iv)^d There is no refutation of $\top$.

From bi-intuitionistic to bi-connexive logic: changing (proof) refutation conditions of (co-)implication

2Int

$$\begin{array}{c}
 ([A], \Gamma; \Delta) \\
 \vdots \\
 \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \overline{A} \quad \overline{A \rightarrow^+ B} \\
 \hline
 B \rightarrow^+ E_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \overline{A} \quad \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_r
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{A \rightarrow^+ B} \\
 \hline
 A \rightarrow^+ E_{r1}
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{A \rightarrow^+ B} \\
 \hline
 B \rightarrow^+ E_{r2}
 \end{array}$$

$$\begin{array}{c}
 (\Gamma; \Delta, \llbracket A \rrbracket) \\
 \vdots \\
 \overline{B} \\
 \hline
 A \rightarrow^- B \rightarrow^- I_r
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \overline{A \rightarrow^- B} \quad \overline{A} \\
 \hline
 B \rightarrow^- E_r
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma'; \Delta') \quad (\Gamma; \Delta) \\
 \vdots \quad \vdots \\
 \overline{A} \quad \overline{B} \\
 \hline
 A \rightarrow^- B \rightarrow^- I_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{A \rightarrow^- B} \\
 \hline
 B \rightarrow^- E_{p1}
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{A \rightarrow^- B} \\
 \hline
 A \rightarrow^- E_{p2}
 \end{array}$$

From bi-intuitionistic to bi-connexive logic: changing (proof) refutation conditions of (co-)implication

2C (Wansing 2016)

$$\begin{array}{c}
 ([A], \Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{B} \\
 \hline
 A \rightarrow^+ B
 \end{array}
 \rightarrow^+ I_p
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A} \quad A \rightarrow^+ B \\
 \hline
 B
 \end{array}
 \rightarrow^+ E_p
 \quad
 \begin{array}{c}
 ([A], \Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{\overline{B}} \\
 \hline
 A \rightarrow^+ B
 \end{array}
 \rightarrow^+ I_r
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A} \quad \overline{\overline{A \rightarrow^+ B}} \\
 \hline
 B
 \end{array}
 \rightarrow^+ E_r$$

$$\begin{array}{c}
 (\Gamma; \Delta, \llbracket A \rrbracket) \\
 \vdots \\
 \hline
 \overline{\overline{B}} \\
 \hline
 A \rightarrow^- B
 \end{array}
 \rightarrow^- I_r
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{\overline{A \rightarrow^- B}} \quad \overline{A} \\
 \hline
 B
 \end{array}
 \rightarrow^- E_r
 \quad
 \begin{array}{c}
 (\Gamma; \Delta, \llbracket A \rrbracket) \\
 \vdots \\
 \hline
 \overline{B} \\
 \hline
 A \rightarrow^- B
 \end{array}
 \rightarrow^- I_p
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A \rightarrow^- B} \quad \overline{\overline{A}} \\
 \hline
 B
 \end{array}
 \rightarrow^- E_p$$

N4 & C vs. 2Int & 2C:

N4

$$\frac{A \quad \sim B}{\sim(A \rightarrow B)} \sim \rightarrow_I \frac{\sim(A \rightarrow B)}{A} \sim \rightarrow_{E_1} \frac{\sim(A \rightarrow B)}{\sim B} \sim \rightarrow_{E_2}$$

C

$$\frac{\begin{array}{c} [A] \\ \vdots \\ \overline{\sim B} \end{array}}{\sim(A \rightarrow B)} \sim \rightarrow_I \frac{A \quad \sim(A \rightarrow B)}{\sim B} \sim \rightarrow_E$$

2Int

$$\begin{array}{ccc} (\Gamma; \Delta)(\Gamma'; \Delta') & (\Gamma; \Delta) & (\Gamma; \Delta) \\ \vdots & \vdots & \vdots \\ \overline{\overline{A}} & \overline{\overline{A \rightarrow B}} & \overline{\overline{A \rightarrow B}} \\ \overline{\overline{B}} & \overline{\overline{A}} & \overline{\overline{B}} \\ \overline{\overline{A \rightarrow B}} & \overline{\overline{A}} & \overline{\overline{B}} \\ \overline{\overline{A \rightarrow B}} \rightarrow_{I^d} & \overline{\overline{A \rightarrow B}} \rightarrow_{E_1^d} & \overline{\overline{B}} \rightarrow_{E_2^d} \end{array}$$

2C

$$\begin{array}{ccc} ([A], \Gamma; \Delta) & (\Gamma; \Delta) & (\Gamma'; \Delta') \\ \vdots & \vdots & \vdots \\ \overline{\overline{B}} & \overline{\overline{A}} & \overline{\overline{A \rightarrow B}} \\ \overline{\overline{A \rightarrow B}} & \overline{\overline{A}} & \overline{\overline{B}} \\ \overline{\overline{A \rightarrow B}} \rightarrow_{I^d} & \overline{\overline{A \rightarrow B}} \rightarrow_{E^d} & \overline{\overline{B}} \end{array}$$

Strong negation in N4, C, 2Int & 2C:

N4

$$\frac{A}{\sim\sim A} \sim\sim I \quad \frac{\sim\sim A}{A} \sim\sim E$$

C

$$\frac{A}{\sim\sim A} \sim\sim I \quad \frac{\sim\sim A}{A} \sim\sim E$$

2Int

2C

$(\Gamma; \Delta)$ $\vdots$ $\frac{\overline{\overline{A}}}{\sim A} \sim I$	$(\Gamma; \Delta)$ $\vdots$ $\frac{\sim A}{A} \sim E$	$(\Gamma; \Delta)$ $\vdots$ $\frac{\overline{\overline{A}}}{\sim A} \sim I^d$	$(\Gamma; \Delta)$ $\vdots$ $\frac{\sim A}{A} \sim E^d$
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⇒ Strong negation serves as toggle between proofs and refutations

Systems C and 2C - Contradictory logics

Contradictory Logics (Niki 2024, Niki & Wansing 2023, Wansing 2024)

Non-trivial negation inconsistent logics:

- Go beyond paraconsistency
- Contain contradictory formulas (in unilateralism: formulas of the form A and $\sim A$) in their set of theorems

Bilateralist contradictory logics (Ayhan 2026):

Non-trivial systems in which there are formulas A that are both provable and refutable.

Example for provable contradiction in C:

$$\frac{\frac{[A \wedge \sim A]}{A} \wedge E_1}{(A \wedge \sim A) \rightarrow A} \rightarrow I \qquad \frac{\frac{[A \wedge \sim A]}{\sim A} \wedge E_2}{\sim ((A \wedge \sim A) \rightarrow A)} \sim \rightarrow I$$

Do we need coordination principles?
Do we need a toggle negation?

(In)Dependence of the two primitive concepts

Do we need horizontal harmony?

In continuation to Gentzen's remark to the effect that (unilateral) I-rules are selfjustifying, and E-rules are justified by the I-rules by the vertical inversion principle, one can see the positive I-rules as a definition that should justify also the negative I-rules. To that end, I introduce a horizontal inversion principle, the definition of which is based on the form of the positive I-rules.

(Francez 2018, p. 1656)

Do we need a toggle negation?

How is the need to add a primitive toggle negation to the system motivated?

- It's nice to have because it makes it easy to switch happily between proof and refutation
 - Is it necessary, though? Is that always desirable?
- ⇒ It creates a very tight relationship between proofs and refutations that may not be justified in all contexts (Ayhan 2025)

$$\frac{\overline{\text{Woman}(t)}}{\sim\text{Woman}(t)}$$

$$\frac{\overline{\text{White}(t)}}{\sim\text{White}(t)}$$

Kick out toggle negation

Argument to consider negation-free systems is not so much about why we *should* get rid of a primitive strong negation but rather: there is no necessity to add this negation to our system and thus, it is a legitimate endeavor to analyze what happens if we go with a negation-free fragment (Wansing, Niki, Drobyshevich 2025, Ayhan 2026)

Still, some motivation:

- Exploring bilateralism with toggle negation seems moot
- ⇒ Bilateral systems with toggle negation are exactly those which can be reduced to unilateral ones
- If negation is understood as refutation, then there's an imbalance: No connective expressing proof

Uniqueness and bilateralism

Uniqueness: the overlooked condition

Introduced by Belnap (1962) in his famous response to Prior's tonk-problem:

What requirements must rules for connectives meet in order to be meaning-conferring?

Belnap's solution: We have to demand

- 1 Conservativity of extensions
- ⇒ 'existence' issue of connectives
- 2 Uniqueness

Characterization of uniqueness

When is a logical connective uniquely determined by its rules?

- If the rules defined permit *at most one* inferential role for the connective to play
- A connective $\#$ is uniquely determined iff when duplicated by a copy-cat connective $\#'$ governed by exactly the same rules, they play exactly the same role, both in premises and in conclusion

How can we check whether a logical connective is uniquely determined by its rules?

$\Rightarrow$ Interderivability qualifying criterion: we must be able to show that for all A, B : $A \# B \vdash A \#' B$ and $A \#' B \vdash A \# B$

Uniqueness: some examples

Conjunction unproblematic

⇒ We can show $A \wedge B \dashv\vdash A \wedge' B$ ($\wedge'$ being defined by the same I- and E-rules as $\wedge$)

$$\frac{\frac{A \wedge' B}{A} \wedge' E \quad \frac{A \wedge' B}{B} \wedge' E}{A \wedge B} \wedge I \qquad \frac{\frac{A \wedge B}{A} \wedge E \quad \frac{A \wedge B}{B} \wedge E}{A \wedge' B} \wedge I'$$

Uniqueness: some examples

Belnap's counterexample: `plonk` and `plink`

Consider two connectives both specified by solely one rule:

- $B \vdash A \text{ plonk } B$
- $B \vdash A \text{ plink } B$

Given only these rules for `plonk`, we cannot show that it is unique:

- Belnap assumes Gentzen's structural rules of reflexivity, weakening, permutation, contraction, and transitivity
- ⇒ Uniqueness iff $A \text{ plonk } B \dashv\vdash A \text{ plink } B$
- ⇒ However: this is impossible to show although `plink` is governed by the same rule

Uniqueness in a bilateralist setting

Test for uniqueness (to within equivalence or synonymy) is some kind of interderivability

⇒ We can show $(A \wedge' B; \emptyset) \vdash^+ A \wedge B$ and $(A \wedge B; \emptyset) \vdash^+ A \wedge' B$ ($\wedge'$ being defined by the same proof rules as $\wedge$)

$$\frac{\frac{\overline{(A; \emptyset) \vdash^+ A} \quad Rf^+}{(A \wedge' B; \emptyset) \vdash^+ A} \wedge' L^a \quad \frac{\overline{(B; \emptyset) \vdash^+ B} \quad Rf^+}{(A \wedge' B; \emptyset) \vdash^+ B} \wedge' L^a}{(A \wedge' B; \emptyset) \vdash^+ A \wedge B} \wedge R^+$$

$$\frac{\frac{\overline{(A; \emptyset) \vdash^+ A} \quad Rf^+}{(A \wedge B; \emptyset) \vdash^+ A} \wedge L^a \quad \frac{\overline{(B; \emptyset) \vdash^+ B} \quad Rf^+}{(A \wedge B; \emptyset) \vdash^+ B} \wedge L^a}{(A \wedge B; \emptyset) \vdash^+ A \wedge' B} \wedge' R^+$$

Uniqueness in a bilateralist setting

⇒ Likewise, we can show $(\emptyset; A \wedge'' B) \vdash^- A \wedge B$ and $(\emptyset; A \wedge B) \vdash^- A \wedge'' B$ ($\wedge''$ being defined by the same refutation rules as $\wedge$)

$$\frac{\frac{\overline{(\emptyset; A) \vdash^- A} \quad Rf^-}{(\emptyset; A) \vdash^- A \wedge'' B} \wedge'' R_1^- \quad \frac{\overline{(\emptyset; B) \vdash^- B} \quad Rf^-}{(\emptyset; B) \vdash^- A \wedge'' B} \wedge'' R_2^-}{(\emptyset; A \wedge B) \vdash^- A \wedge'' B} \wedge L^c$$

$$\frac{\frac{\overline{(\emptyset; A) \vdash^- A} \quad Rf^-}{(\emptyset; A) \vdash^- A \wedge B} \wedge R_1^- \quad \frac{\overline{(\emptyset; B) \vdash^- B} \quad Rf^-}{(\emptyset; B) \vdash^- A \wedge B} \wedge R_2^-}{(\emptyset; A \wedge'' B) \vdash^- A \wedge B} \wedge'' L^c$$

Uniqueness-criterion for the bilateralist setting

Definition of uniqueness for bilaterally defined connectives (Ayhan 2021):

In a bilateralist setting with consequence relations for verification as well as falsification, two n -place connectives $\#$ and $\#'$, which are defined by exactly the same set of inference rules, play exactly the same inferential role, i.e. are unique, iff for all $A_1, \dots, A_n$ the formulas $\#(A_1, \dots, A_n)$ and $\#'(A_1, \dots, A_n)$ are interderivable **as well as dually interderivable**. To express this formally for the case of 2Int:

- (i) $(A \# B; \emptyset) \vdash^+ A \#' B$ and $(A \#' B; \emptyset) \vdash^+ A \# B$
- (ii) $(\emptyset; A \# B) \vdash^- A \#' B$ and $(\emptyset; A \#' B) \vdash^- A \# B$.

Is negation unique in FDE?

Uniqueness of negation in FDE

The unilateral natural deduction rules $\mathcal{R}_{\mathbf{FDE}}$ for **FDE** are all the following rules:

$$\begin{array}{ccccccc}
 \frac{A \quad B}{A \wedge B} \wedge I & \frac{A \wedge B}{A} \wedge E & \frac{A \wedge B}{B} \wedge E & \frac{A}{A \vee B} \vee I & \frac{B}{A \vee B} \vee I & \frac{A \vee B \quad \begin{array}{c} [A] \\ \vdots \\ C \end{array} \quad \begin{array}{c} [B] \\ \vdots \\ C \end{array}}{C} \vee E \\
 & & \frac{A}{\neg\neg A} \neg\neg I & \frac{\neg\neg A}{A} \neg\neg E & & & \\
 \frac{\neg A \vee \neg B}{\neg(A \wedge B)} \neg\wedge I & \frac{\neg(A \wedge B)}{\neg A \vee \neg B} \neg\wedge E & \frac{\neg A \wedge \neg B}{\neg(A \vee B)} \neg\vee I & \frac{\neg(A \vee B)}{\neg A \wedge \neg B} \neg\vee E & & &
 \end{array}$$

Or

$$\frac{\neg A \quad \neg B}{\neg(A \vee B)} \neg\vee I \quad \frac{\neg(A \vee B)}{\neg A} \neg\vee E \quad \frac{\neg(A \vee B)}{\neg B} \neg\vee E \quad \frac{\neg A}{\neg(A \wedge B)} \neg\wedge I \quad \frac{\neg B}{\neg(A \wedge B)} \neg\wedge I \quad \frac{\neg(A \wedge B) \quad \begin{array}{c} [\neg A] \\ \vdots \\ C \end{array} \quad \begin{array}{c} [\neg B] \\ \vdots \\ C \end{array}}{C} \neg\wedge E$$

Failure of uniqueness of negation in unilateral FDE

We create a copy-cat connective for negation:

$$\frac{A}{\neg' \neg' A} \neg' \neg' I \quad \frac{\neg' \neg' A}{A} \neg' \neg' E$$

$$\frac{\neg' A \quad \neg' B}{\neg' (A \vee B)} \quad \frac{\neg' (A \vee B)}{\neg' A} \quad \frac{\neg' (A \vee B)}{\neg' B} \quad \frac{\neg' A}{\neg' (A \wedge B)} \quad \frac{\neg' B}{\neg' (A \wedge B)} \quad \frac{\neg' (A \wedge B) \quad \begin{array}{c} [\neg' A] \\ \vdots \\ C \end{array} \quad \begin{array}{c} [\neg' B] \\ \vdots \\ C \end{array}}{C}$$

Can you show $\neg A \dashv\vdash \neg' A$ for all A ?

No. (Humberstone 2019)

In bilateral setting? Ezpz. (Ayhan, Toyooka & Omori forthcoming)

Uniqueness of negation in bilateral FDE

$$\begin{array}{cc}
 \begin{array}{c} (\Gamma; \Delta) \\ \vdots \\ \overline{\overline{A}} \\ \hline \neg' A \end{array} \neg' I^+ &
 \begin{array}{c} (\Gamma; \Delta) \\ \vdots \\ \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg' E^+ \\
 \\
 \begin{array}{c} \overline{\neg A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg E^+ &
 \begin{array}{c} \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg' E^+ \\
 \hline
 \begin{array}{c} \overline{\neg A} \\ \hline \overline{\overline{A}} \\ \hline \neg' A \end{array} \neg' I^+ &
 \begin{array}{c} \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline \neg A \end{array} \neg I^+
 \end{array}
 \qquad
 \begin{array}{cc}
 \begin{array}{c} (\Gamma; \Delta) \\ \vdots \\ \overline{\overline{A}} \\ \hline \neg' A \end{array} \neg' I^- &
 \begin{array}{c} (\Gamma; \Delta) \\ \vdots \\ \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg' E^- \\
 \\
 \begin{array}{c} \overline{\neg A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg E^- &
 \begin{array}{c} \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline A \end{array} \neg' E^- \\
 \hline
 \begin{array}{c} \overline{\neg A} \\ \hline \overline{\overline{A}} \\ \hline \neg' A \end{array} \neg' I^- &
 \begin{array}{c} \overline{\neg' A} \\ \hline \overline{\overline{A}} \\ \hline \neg A \end{array} \neg I^-
 \end{array}$$

Summary

You can do weird and cool stuff with bilateralism.

Thanks for your attention!

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What could you look into?

Lots of stuff!

- Connexive logics
- Truth/Falsemaker semantics (something we didn't mention at all)
- Multilateralism
- Meta-inferential bilateralism
- ...?

Time for Questions and Feedback!